

One-Way ANOVA — Part B

Within & Between Group Variability Student Worksheet

Name: _____

Date: April ____, 2026

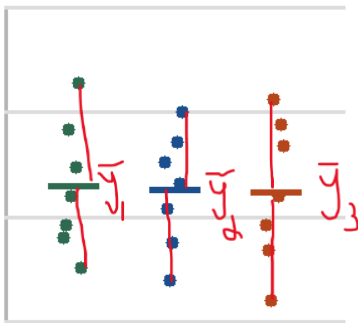
Context: This section builds conceptual understanding of the two sources of variability that one-way ANOVA uses to compare group means: **within-group variability** (noise) and **between-group variability** (signal). The response variable (y) is plotted vertically in all dot-plots.

1 Within-Group Variability

Within-group variability measures how spread-out individual response variable values are around their own group mean $\bar{y}_i$. It is the **noise** in ANOVA — variation that exists even within groups receiving the same treatment.

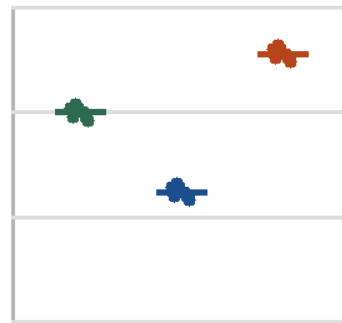
The dot-plots below show the response variable (y = exam score) on the **vertical axis** and groups (A, B, C) on the horizontal axis.

High Within-Group Variability



Dots spread far within each group

Low Within-Group Variability



Dots tight within each group

1a Fill in the blanks

When within-group variability is **large**, individual response values are spread **far** from their group mean $\bar{y}_i$.

This makes it **HARD** for ANOVA to detect real differences between groups, because

the **noise** grows larger.

1b Identify which plot shows larger within-group variability and explain why it matters

The left plot has the higher within group variability because the points dramatically deviate from each groups mean. When the within group variability spread is large, the one-way ANOVA cannot detect the real differences between the groups means.

2 Between-Group Variability

Between-group variability measures how far apart the **group means $\bar{y}_i$** are from the overall mean **$\bar{\bar{y}}$** (the grand mean across all observations). It is the **signal** — evidence that the groups actually differ.

Using the data from the study: group means are $\bar{y}_A = 72$, $\bar{y}_B = 85$, $\bar{y}_C = 91$, and the overall mean is $\bar{\bar{y}} = 83$.

2a Calculate deviations

Group	Group Mean $\bar{y}_i$	Overall Mean $\bar{\bar{y}}$	Deviation $\bar{y}_i - \bar{\bar{y}}$
Group A	72	83	$72 - 83 = -11$ points
Group B	85	83	$85 - 83 = 2$ points
Group C	91	83	$91 - 83 = 8$ points

2b Interpretation

The group with the largest deviation from the overall mean is contributing most to between-group variability. Which group is it, and what does this suggest about the effectiveness of that study method?

Group A is deviating the most from the overall mean with 11 point below and group C is 8 points above. The lowest performing group is re-reading.

3 The F-Ratio: Signal Over Noise

The **F-statistic** is computed as:

$$F = \frac{\text{Between-Group Variability}}{\text{Within-Group Variability}} = \frac{\text{MSB}}{\text{MSW}}$$

$$\frac{\text{Mean Square Between}}{\text{Mean Square Within}}$$

$$\frac{\text{Squared Deviations on average}}{\text{Squared Deviations on average}}$$

3a Fill in the blanks

When $F \gg 1$, the between-group signal is **large** relative to within-group noise. This **supports** the alternative hypothesis that at least one group mean $\bar{y}_i$ differs from the others.

When $F \approx 1$, the between-group signal is **similar in size** to within-group noise. This is consistent

with **the null hypothesis that the populations means are the same.**

Can F be negative? No, Reason: Since we are square the deviations, the F statistic has to be 0 or more.

3b Interpret $F = 9.14$ (from the ANOVA output table in Part A) means that the between group variability is approximately 9 times larger than within group variability. The strong signal relative to the noise is apparent providing evidence against the null hypothesis.

4 Equal Variance Condition

ANOVA assumes that within-group variability is **roughly equal across all groups** — this is called **homoscedasticity**. If violated, the pooled MSW (denominator of F) is distorted.

4a Check the condition

Group	Scores	Std Dev (approx.)
A	78, 80, 79, 81, 77	~1.6
B	42, 65, 88, 91, 55	~21.1
C	72, 74, 73, 75, 71	~1.6

Ratio of largest to smallest SD: $21.1/1.6 \approx 13$ Is the condition met (rule: largest SD $\leq 2 \times$

smallest)? **No, the assumption has been clearly violated**

Explain what this violation means for interpreting the F-statistic. **-**

